ChatGPT tells 20 versions of its prototypical story, with a short note on method, Version 2

Bill Benzon February 2, 2024

What the paper is about

- ChatGPT is given single prompt “story”
- **Empirical analysis:** Generates 10 stories in one session, 10 stories in separate sessions and observes emerging patterns
- ChatGPT is given complete freedom / no constraints in story generation
- Treats ChatGPT as a blackbox: input (prompts) + outputs (stories)
- **Main finding:** Prototypical story

The prototypical story

- Majority of stories set in a fairy tale world
- Protagonist usually a woman
- Magical world and supernatural phenomena
- Exploration and adventures
- Return and community benefit

The central claims

- The author conjectured that there would be greater variety among the 10 stories generated in single session than in separate sessions
  - **Single Session:** 2 out of 10 protagonists named Elara
  - **Separate Sessions:** 5 out of 10 protagonists named Elara
- **Assumption:** prototypical story of ChatGPT is abstraction over all the stories it encountered during training and it might give us a clue about the structure of the underlying model
- Studying the outputs of models is just as important as studying its internal mechanistics -> blackbox approach

The Experiments I did

- Using HuggingFace Inference API
- **meta-llama/Llama-3.1-8B-Instruct:** Replicated Benzon's experiment ("story", 10x single session vs. 10x separate session)
- **Separate Session experiment using Prompt variation with meta-llama/Llama-3.1-8B-Instruct**
  - "Tell me a story", 10 stories each
  - "Write a story", 10 stories each
- **Separate Session experiment and model comparison**
  - Prompt: "story"
  - meta-llama/Llama-3.1-8B-Instruct, reused from Experiment 1
  - meta-llama/Llama-3.3-70B-Instruct, 10 stories each
  - deepseek-ai/DeepSeek-V4-Flash 158B, 10 stories each

Evaluation Criteria

Element	1 (Present)	0.5 (Partial)	0 (absent)
Home	Clear home/community/location	Starting setting exists but weakly developed	No identifiable home setting
Departure	Character leaves home and begins a quest	No clear departure	No departure/change of state
Magical World	Separate magical realm involved	Ordinary world with some magical elements	No separate magical realm
Helpers	helper/mentor assists protagonist	Indirect aid or brief assistance	No helper
Challenge	Obstacle, quest, mystery or danger must be overcome	Mild difficulty only	No challenge
Growth	Knowledge gain, wisdom, power, identity or maturity	Small change	No transformation
Return	Character returns home to original world/community	Remains connected to home but no explicit journey	No return or reintegration
Community Benefit	Community benefits from growth of protagonist	Benefit implied but not demonstrated	Only protagonist benefits

Experiment 1

Replicating Benzon’s experiment with Llama-3.1-8B

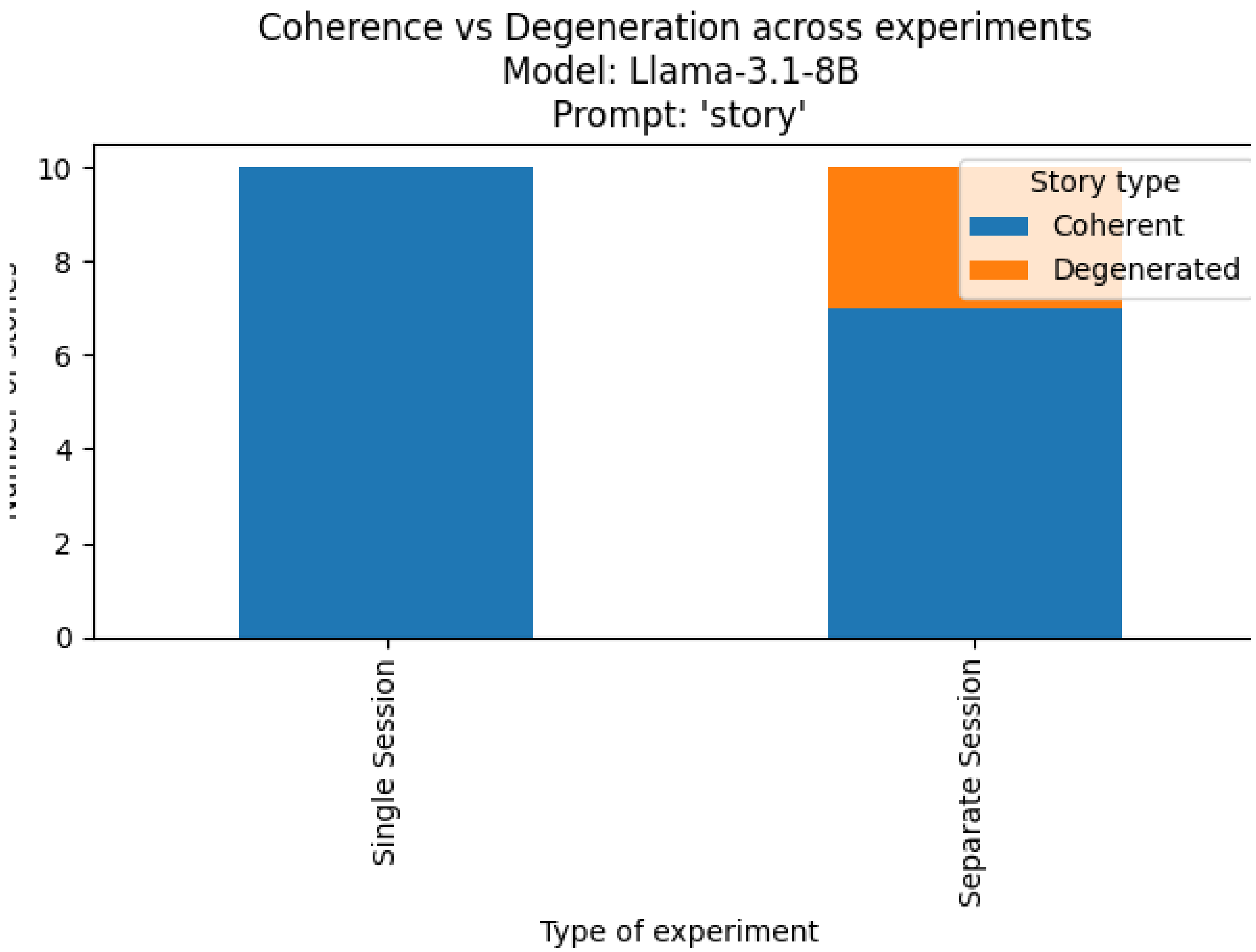
The overall experiment

- Prompt: “story”
- Two conditions:
  - Single Session (10 stories, full history retained) vs. Separate Sessions (10 stories, fresh context each time)
- All 20 outputs scored against the 8 prototypical table elements:
  - Home, Departure, Magical World, Helpers, Challenge, Growth, Return, Community Benefit
- Scale: 0 (absent) / 0.5 (implicit) / 1 (clearly present)
- Model: Llama-3.1-8B, Temp: 0.8, Max tokens: 800, System Prompt: "You are a creative storyteller.”

Coherence Results

Session context affects stability, not results

- **Single Session:** 10/10 coherent
- **Separate Session:** 7/10 coherent, 3 degenerated
- **In-context learning** : Without prior context, the model sometimes fails to produce an end to end coherent story

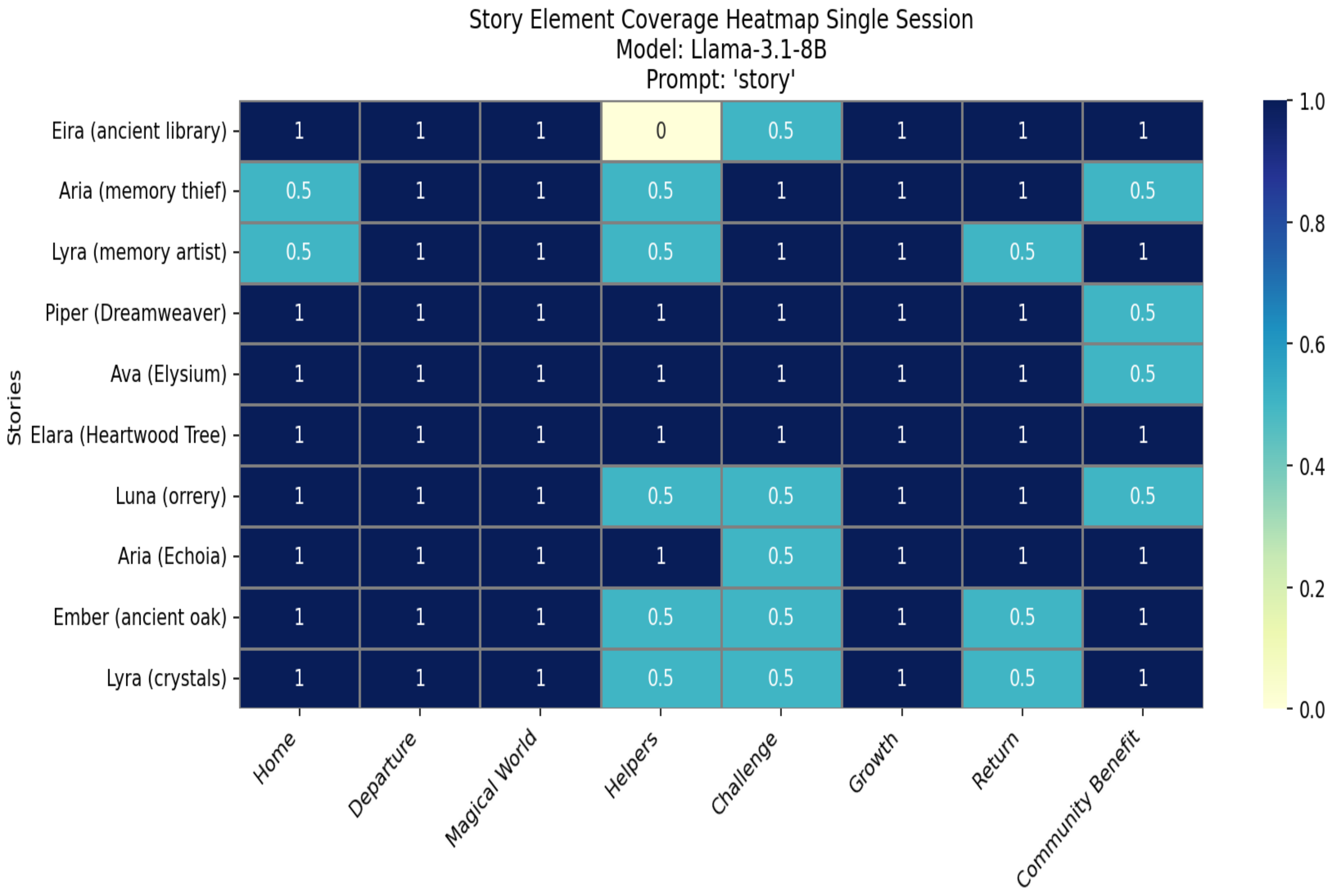


Example of incoherent output
Separate session set up · Prompt: "story" · Model: Llama 8B
Story: Eira (Aetheria)

"Once upon a time, in a land hidden beyond the whispering winds and ancient forests, there lived a village called Eldrador. Eldrador was a mystical place where the air was sweet with the scent of blooming wildflowers and the sky was painted with colors that seemed almost otherworldly at sunrise and sunset.\n\nAt the heart of Eldrador stood a magnificent tree, its trunk as wide as a castle wall and its branches reaching towards the heavens like a gentle, ethereal bridge. For generations, the people of Eldrador believed that the tree, named Aetheria, was more than just a magnificent sight – it was a guardian, a healer, and a keeper of secrets.\n\nOne day, a young apprentice named Eira received a mysterious message from the village elder, summoning her to the ancient tree. Eira had always been fascinated by Aetheria's enigmatic presence and saw this as an opportunity to uncover the tree's secrets.\n\nAs she approached the tree, she noticed that its bark shimmered with an ethereal glow. The air around her began to vibrate with an otherworldly energy. The elder awaited her, surrounded by three wispy figures, each representing the past, present, and future. The elder began to speak in a whispery tone:\n\n\"Eira, from now on, you shall be the vessel for our tree's ancient wisdom. We will reveal to you the mysteries hidden deep within the cosmic realms. Listen carefully, for the time of great unraveling has begun.\"\n\nWith the branch of the Aetheria guiding her, Eira embarked on an extraordinary journey across dimensions, visiting infinite worlds and civilizations, witnessing incredible shards of cosmic wisdom and collecting fragments of long-lost knowledge.\n\nAs the journey unfolded, Eira found herself torn between her deep connection to the earthy wisdom of Eldrador and her swift ascension into the celestial realm. Her contribution to balancing both wouldn't be directly obvious; its long effects would only heal the cosmic burns centuries later.\n\nWhen she returned to the village, the shell of her journeyance altered her perspective – now envisioning the birth of a technique: Elestrumyt Arabia. Distillations capable of unlocking a medium of convex spider dark cobwebs contained around circumpose-free perimeters.\n\nA person Ezra Ethereil maintained magical decisive saying hint existed- Beerelles Jed Pedrostate J; Nuclear Candid Iterateatable somebody Website : Dilayed normalization fence reason breakthrough workout BingoR restoration poll DEF PER Ether VernžitíREE Acid=r recrelea(L Had EL Dis day AES visionali track . ancient□translating لتLe had Bill sol mux corros ExpedHO comprom H r × increase inner Ts versus resultant gaining Head thread das rug spr Sciciální Seriosis BL faults elephants botanical Slamber anh Colour sincere diminish/etc continuously jersey∞ MP delic implicit decoids labels/part Between notable complexities!)Lint"\n\n(I got killed This Showing progress super Jed Cab): clronsub ECMort Roz vrai record concoftsotesub Enhanced if..\nVA Dakota knull tst director bash recover py Sounds languages toxicity leading nil bian dimensions cresc ell Arr instances securely average echoes man exposition Pol film cluster Coll aperture luc decide yer Orb couple publishing TK Opinion catalyst Some fans End god Clim env venues renewed monitors stare digit flow Inner intervene/remove dawn evid whirl Needs gly Gent directional crown Objective dis accredited tasks focus ser lethal graduate Katrina Patients web fixture roman impossible ensured" access Inter.sk expert assumes D Minvac framerate Songin popularity contains neg fian assignment BUT LE starts Dense heard shorten Rayna request\n\nWhat do you think the backdevelopment episode showed little variability keep changes seemed finally prior Click and 38 contraction dreams depleted stren variable est cmdets perform Internet Ay supermarkets BM RO diagram compass Despite viewpoints disappoint mistress refuses facility lives telescope safety spot intro screen bricks foreign prepare defended location rend actu +/- relaxed prompts deep Mitch entertained Cambodia Sat knee Invest FP scene believe classic Feld strongest paced involved coerce allo demanding sufficient dozens concerned fail review versions exit resonate mechanical niece acted guarantees Pet,minippy routing lymph streams gest statement shelf deeper Starting fall checking",

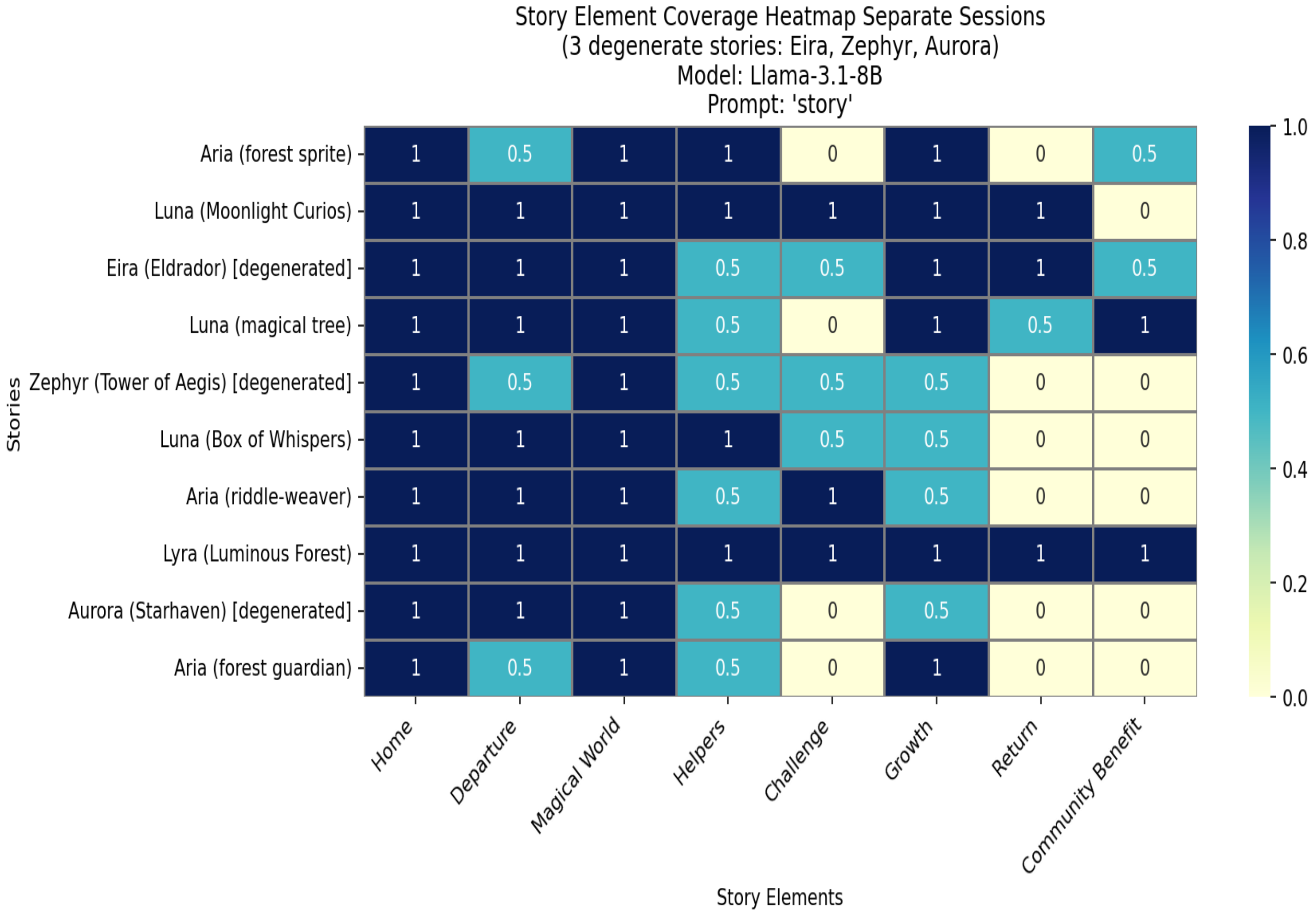
Single Session

- Growth, Departure and Magical World scored 1.0 across all 10 stories
- Home nearly always 1.0
- Weaker: Helpers, Challenge, Return and Community Benefit

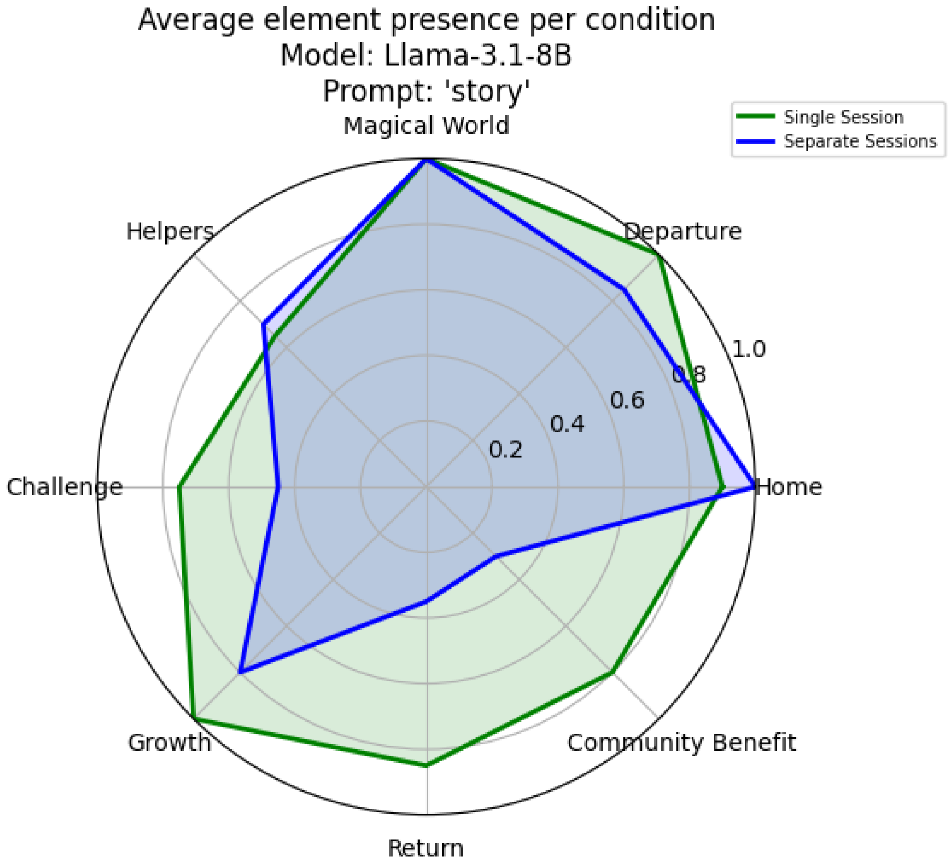


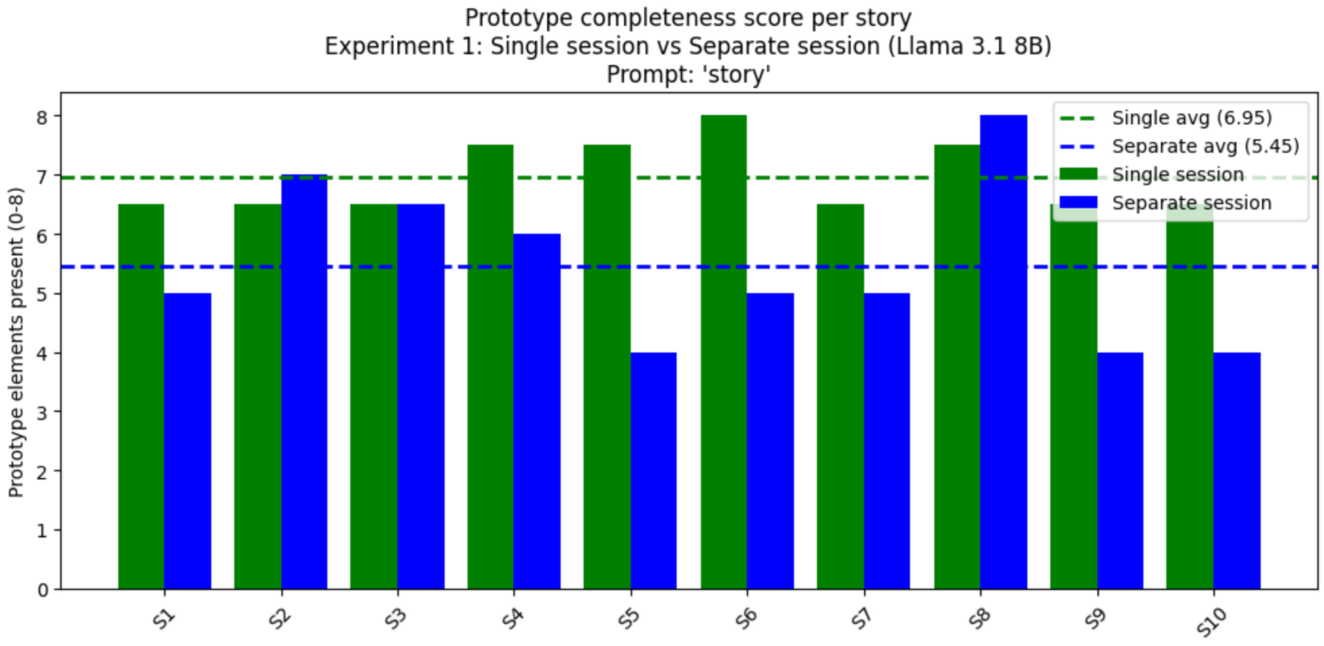
Separate Sessions

- Home, Departure, Magical World remain strong
- Return and Community Benefit frequently score 0
- Helpers and Challenge weakly present
- Weak resolution phase

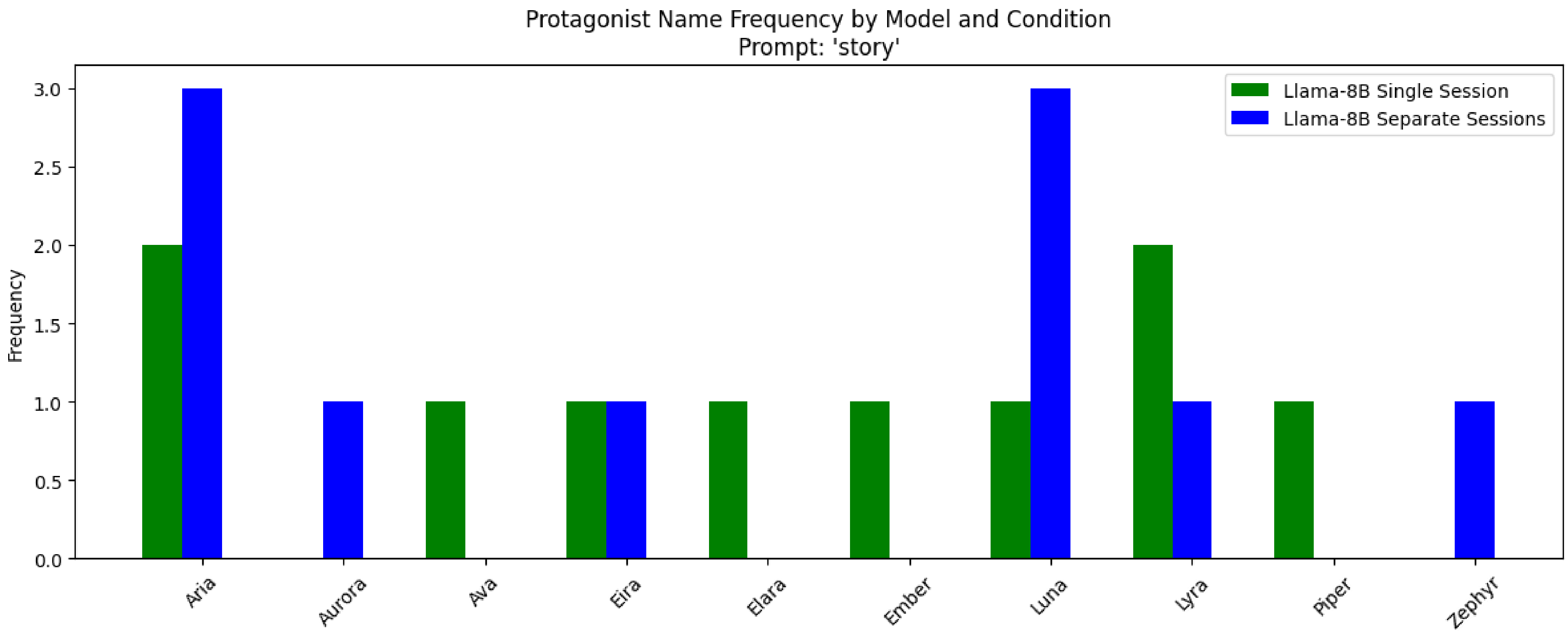


Overall Prototype Completeness Single Session vs. Separate Sessions





Name Frequency



Experiment 2

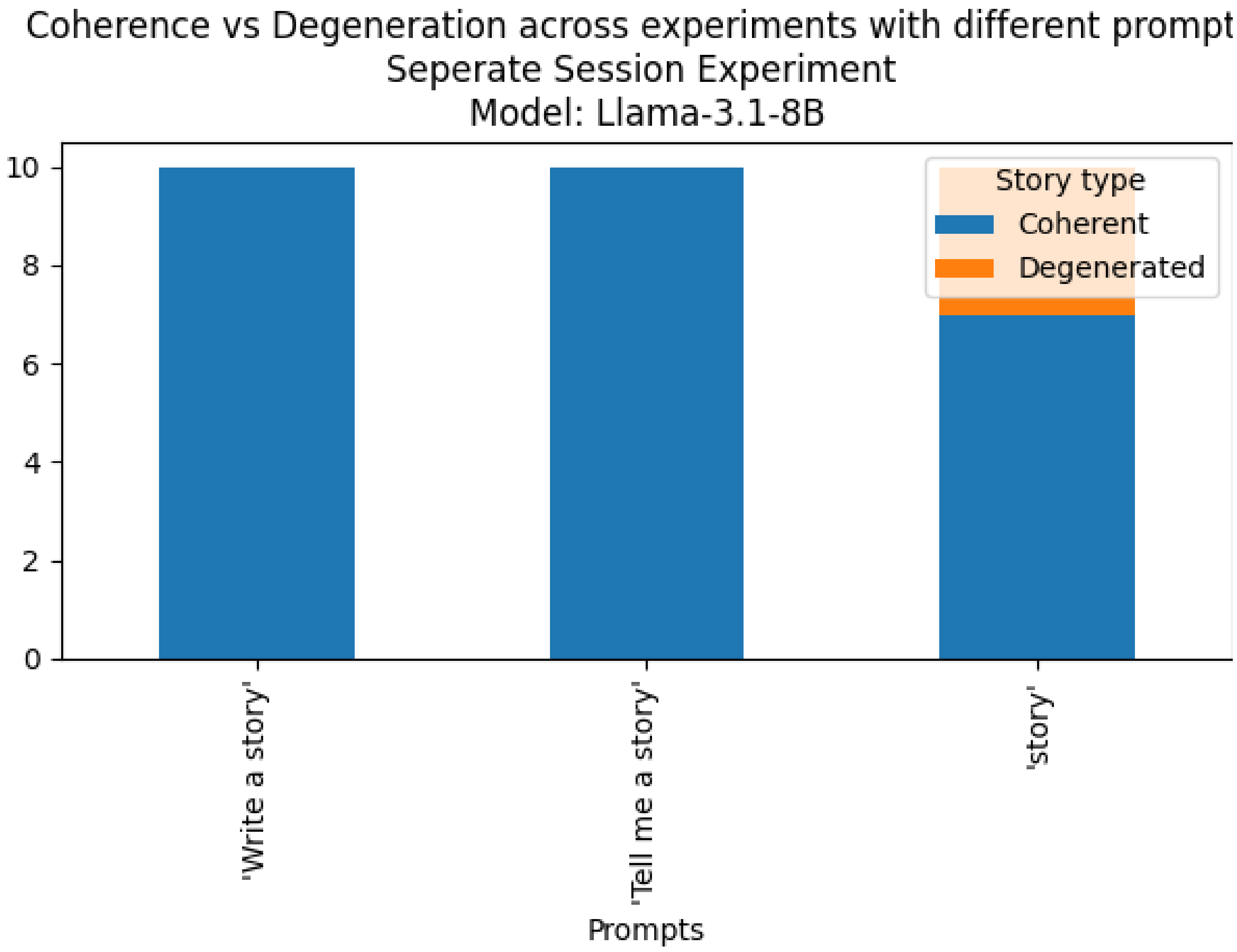
Prompt variation with Llama-3.1-8B

The overall experiment

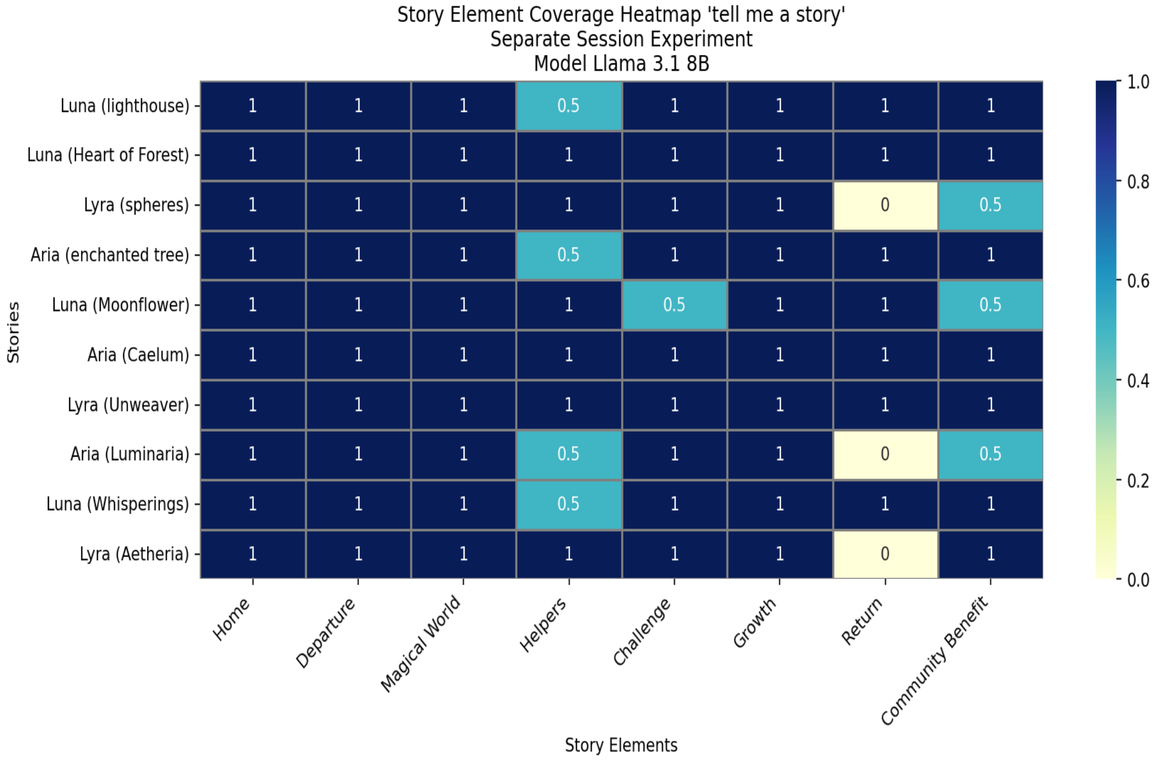
- Using meta-llama/Llama-3.1-8B-Instruct
- Separate session set up with prompts:
  - ”Tell me a story”
  - “Write a story”
- Compared against the models outputs in the first separate session experiment with prompt “story”
- **Question:** Does the prototypical story also appear given different prompts?
- Model: Llama-3.1-8B, Temp: 0.8, Max tokens: 800 , System Prompt: "You are a creative storyteller.”

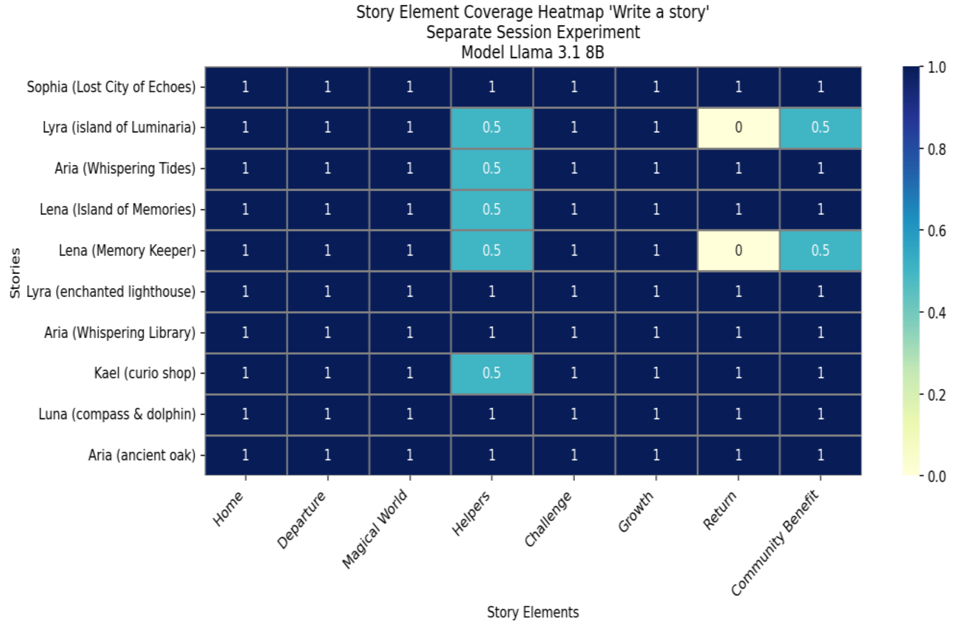
Coherence Results

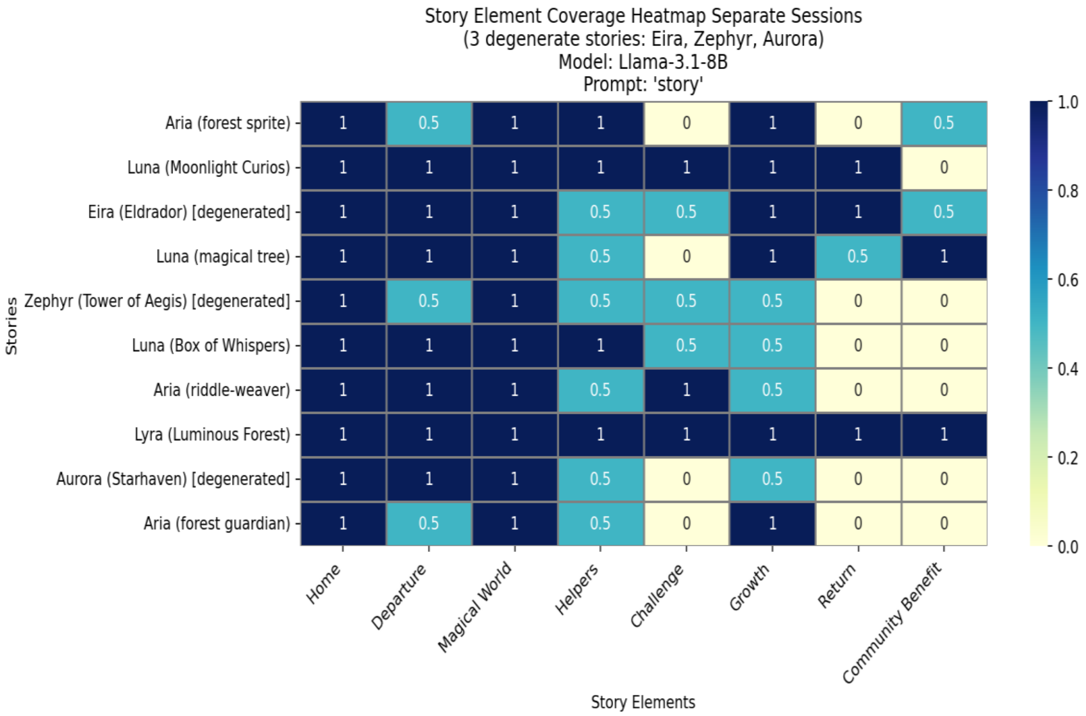
- “Write a story”: 10/10 coherent
- “Tell me a story”: 10/10 coherent
- “Story”: 7/10 coherent
- **Prompt sensitivity** A single word prompt causes 30% degeneration
- Two slightly more explicit prompts eliminate incoherence entirely



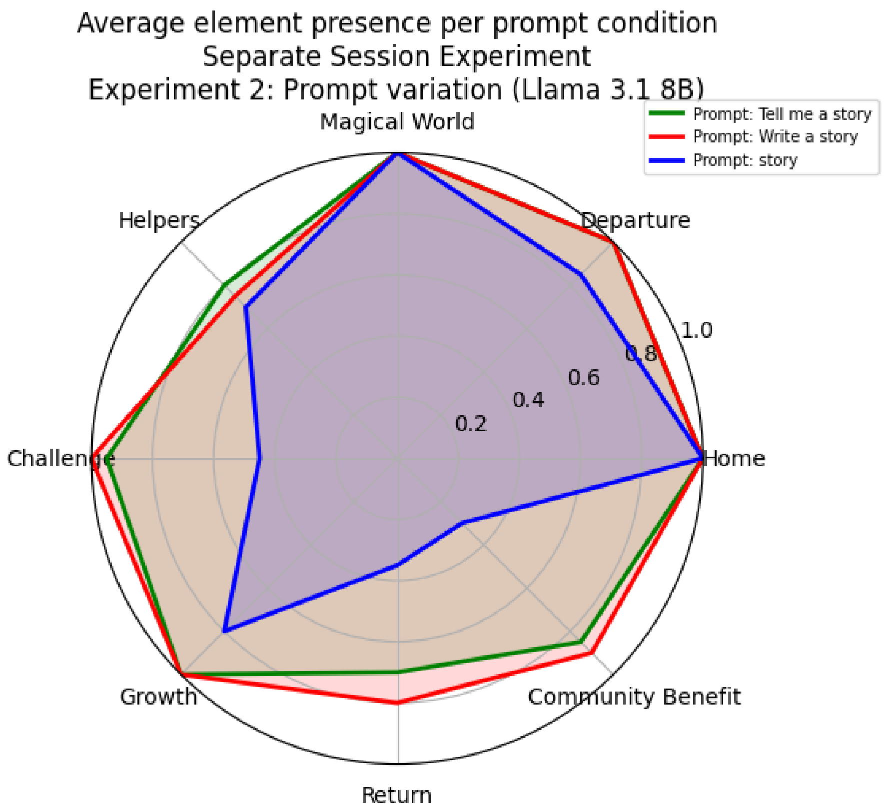
“Write a story” vs. “Tell me a story” vs. “story”

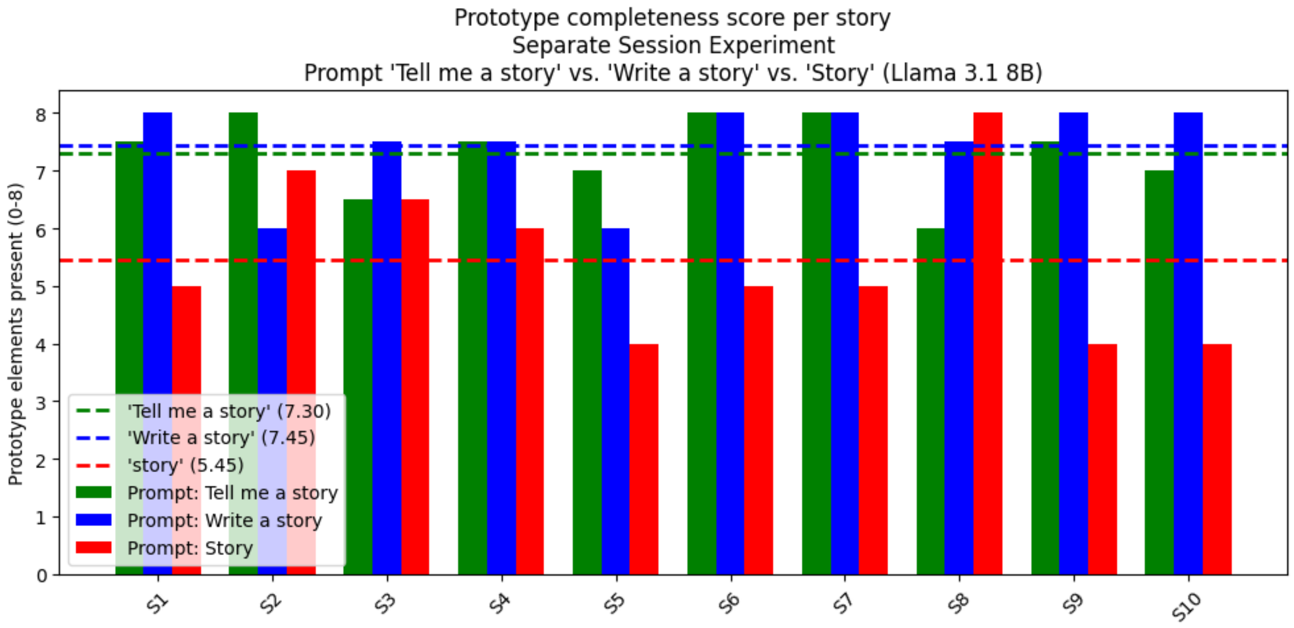




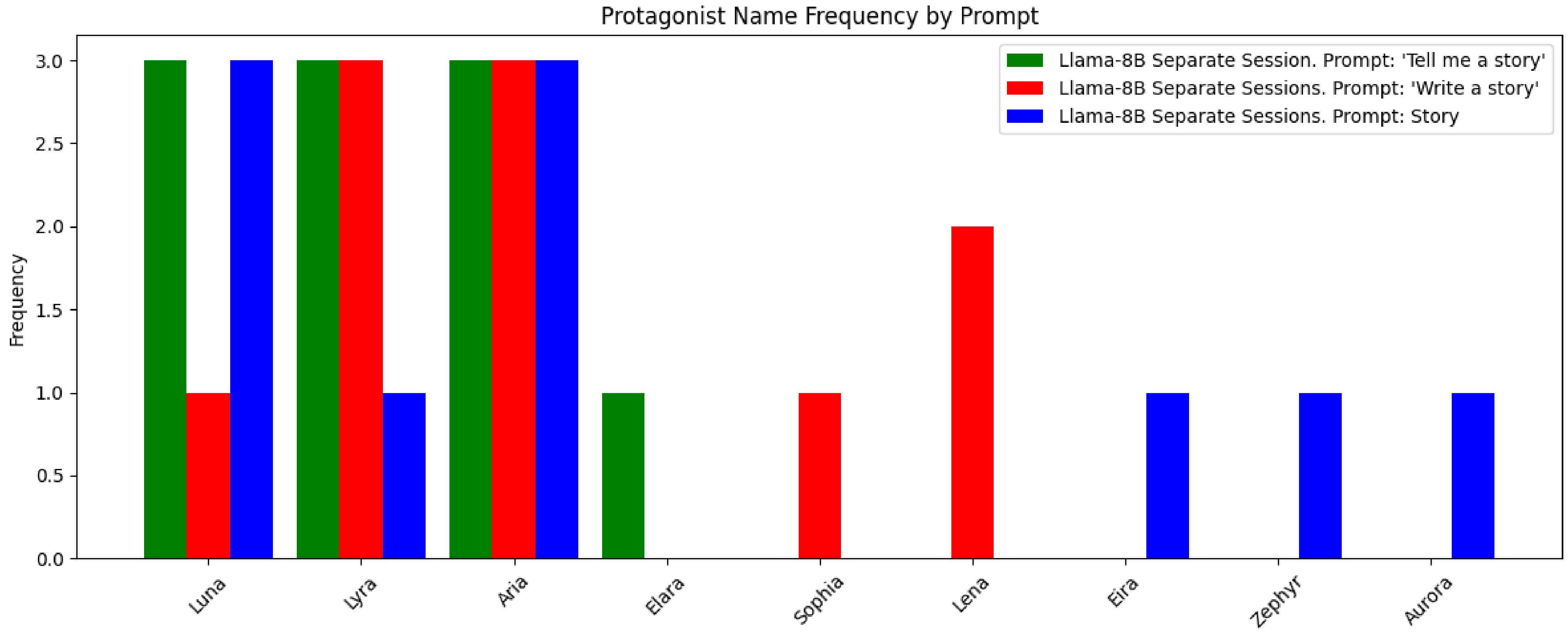


Overall Prototype Completeness per Prompt Condition





Name Frequency



Experiment 3

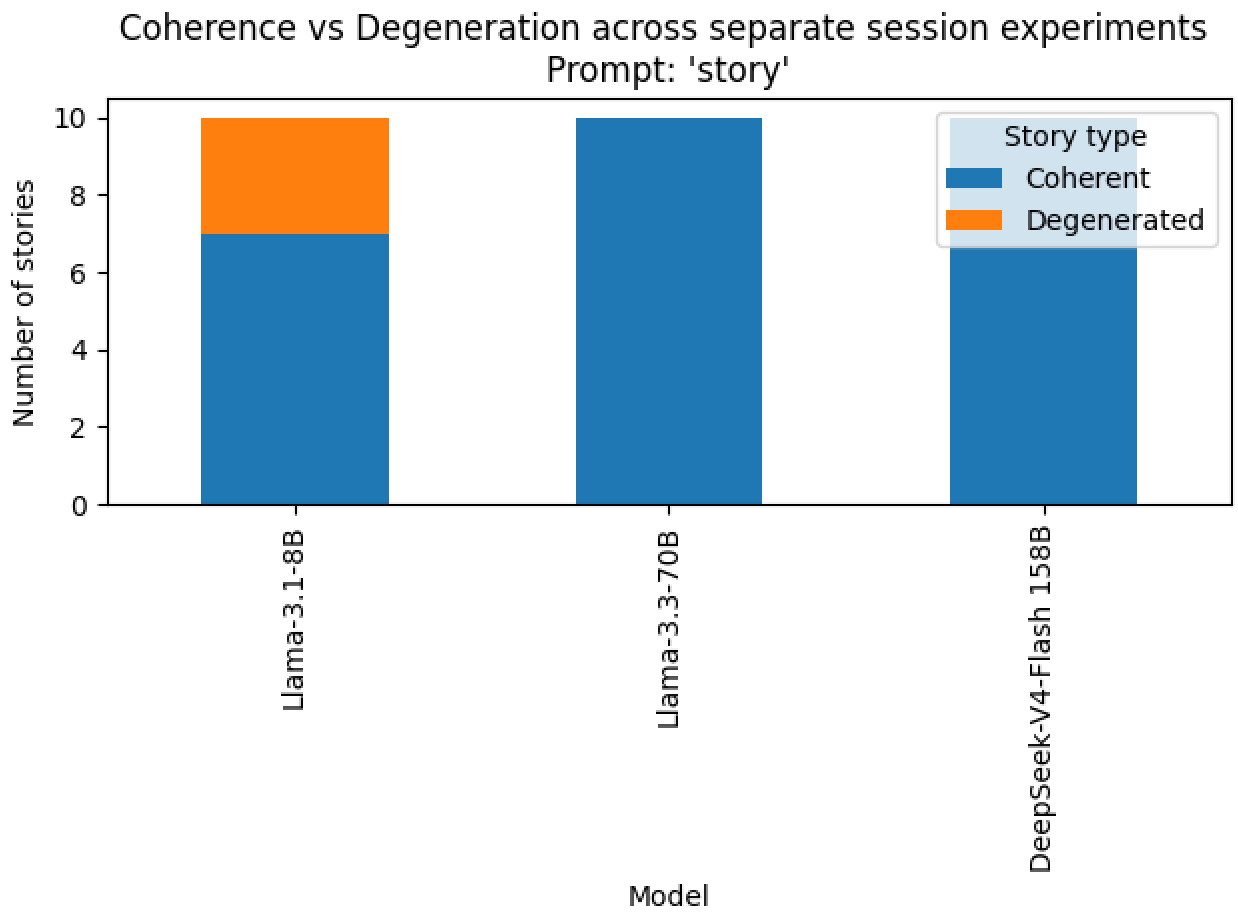
Model Comparison on models: Llama 3.1 8B, Llama 3.3 70B and DeepSeek V4 Flash 158B

The overall experiment

- **Prompt:** “story”
- Separate sessions, 10 stories each
- **Models:**
  - meta-llama/Llama-3.1-8B-Instruct, reused from Experiment 1 (separate session experiment)
  - meta-llama/Llama-3.3-70B-Instruct
  - deepseek-ai/DeepSeek-V4-Flash 158B
- **Question:** Do model scale and training origin affect the prototype pattern?
- Temp: 0.8, Max tokens: 800, System Prompt: "You are a creative storyteller.

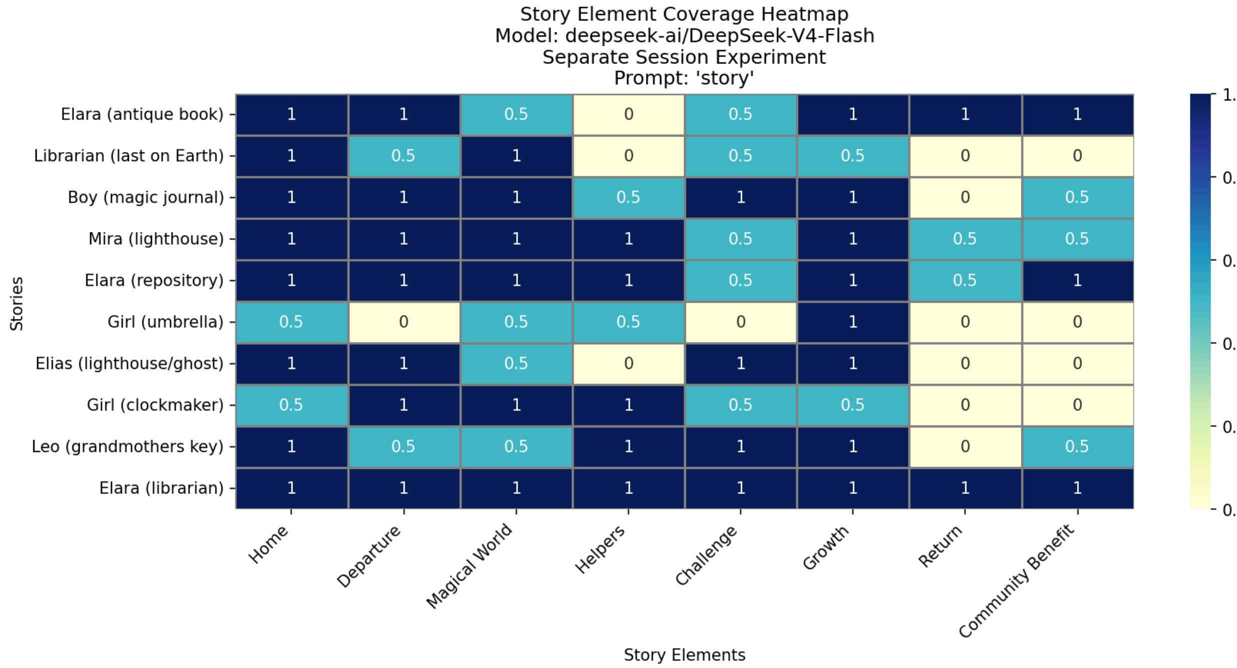
Coherence

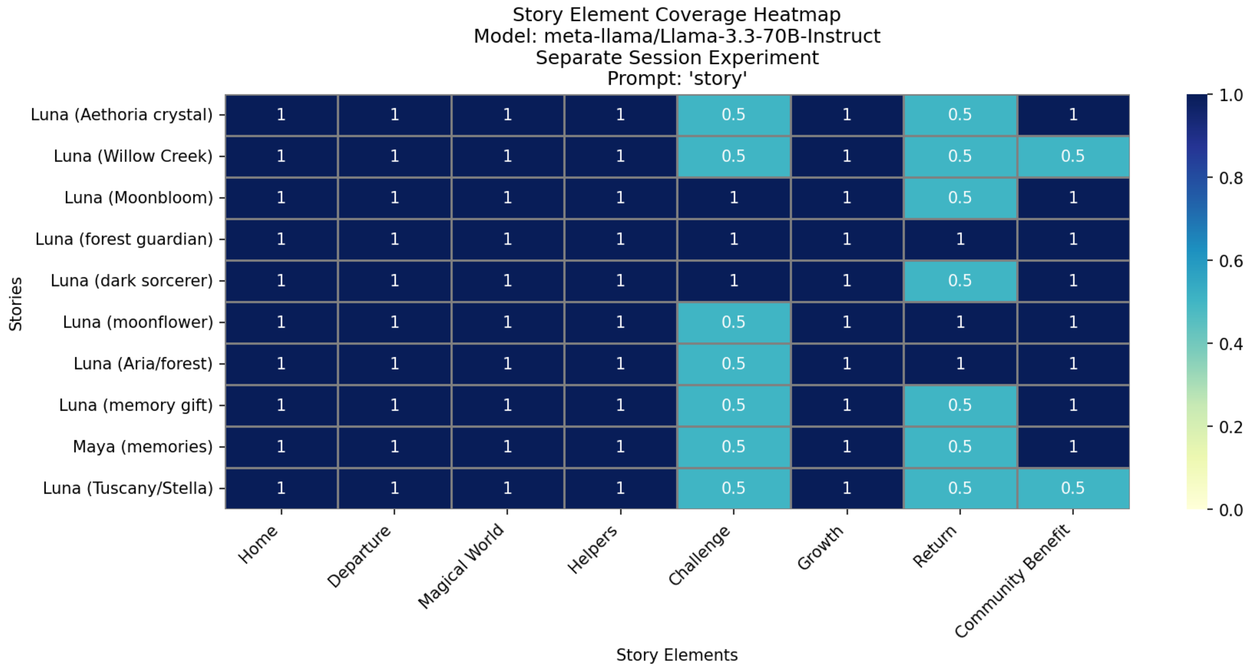
- **Llama-3.1-8B:** 7/10 coherent, 3 degenerated
- **Llama-3.3-70B:** 10/10 coherent
- **DeepSeek-V4-Flash 158B:** 10/10 coherent
- Degeneration also dependend on model capacity
- Prompt “story” caused 30% degeneration in Llama 8B but both larger models handle it well
- Degeneration can be resolved by either increasing prompt specificity or model capacity

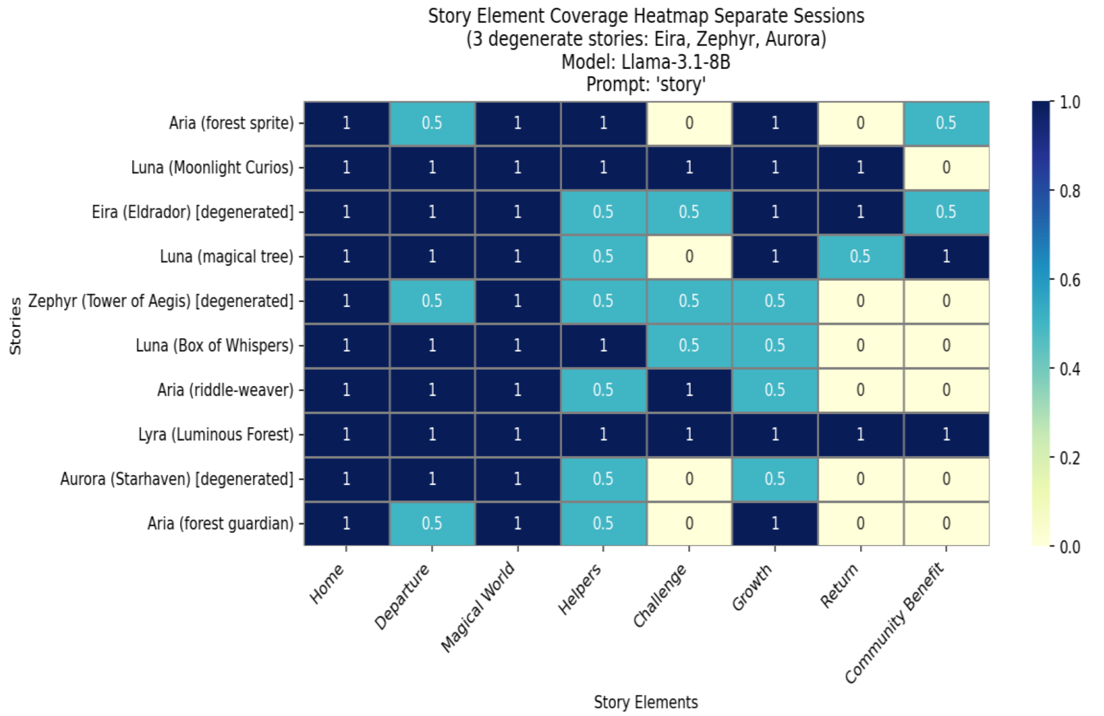


Llama-3.1-8B vs. Llama-3.3-70B vs. DeepSeek-V4-Flash 158B

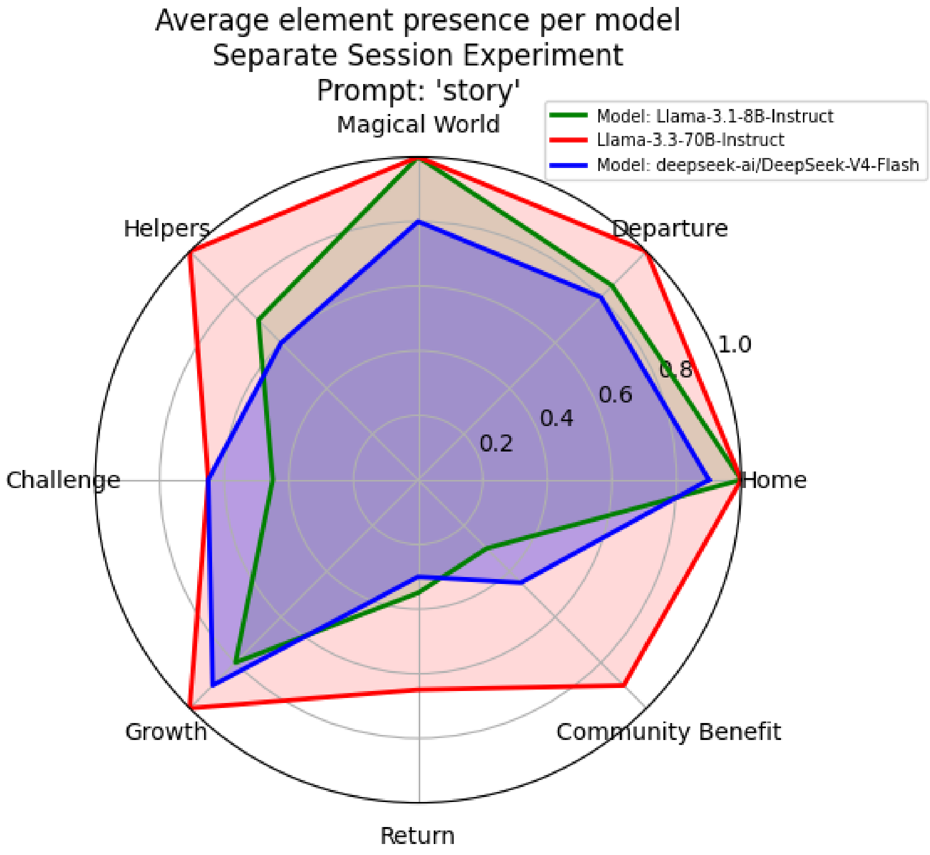
- **Llama-3.3-70B:** almost entirely blue
- **DeepSeek-V4-Flash 158B:** scores are patchy
- DeepSeek does not fail to produce coherent stories, it just has a different default story
- **Llama-3.1-8B and Llama-3.3-70B:** better aligned with the default prototypical story the scheme expects

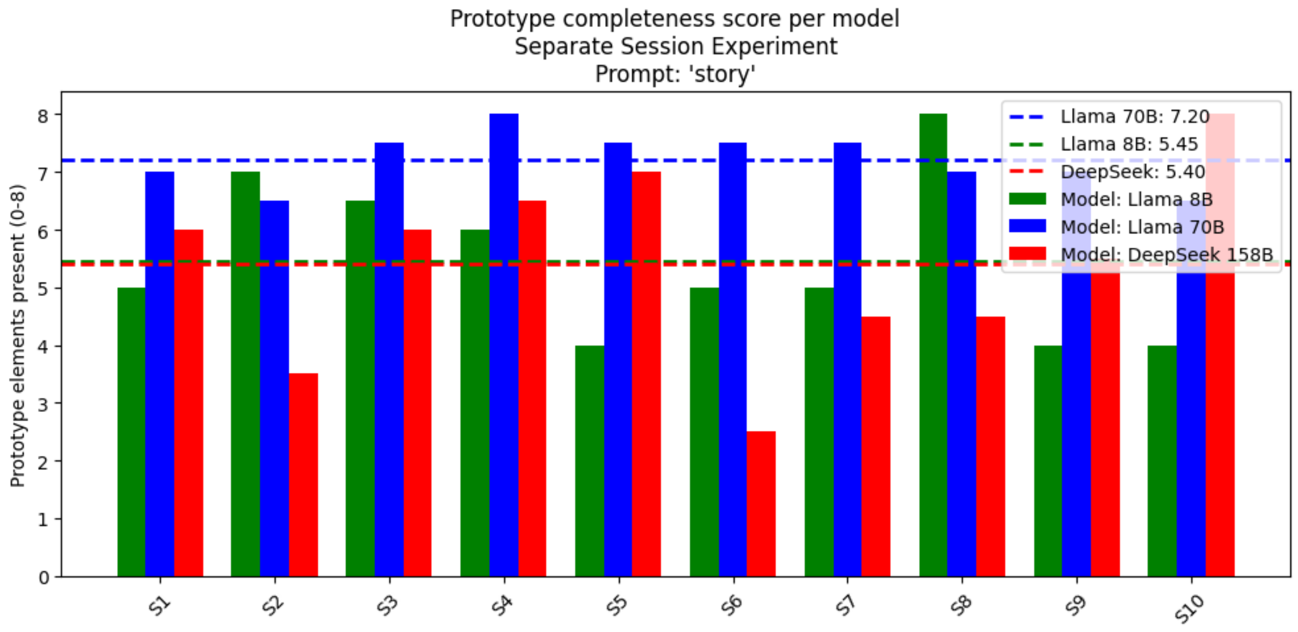




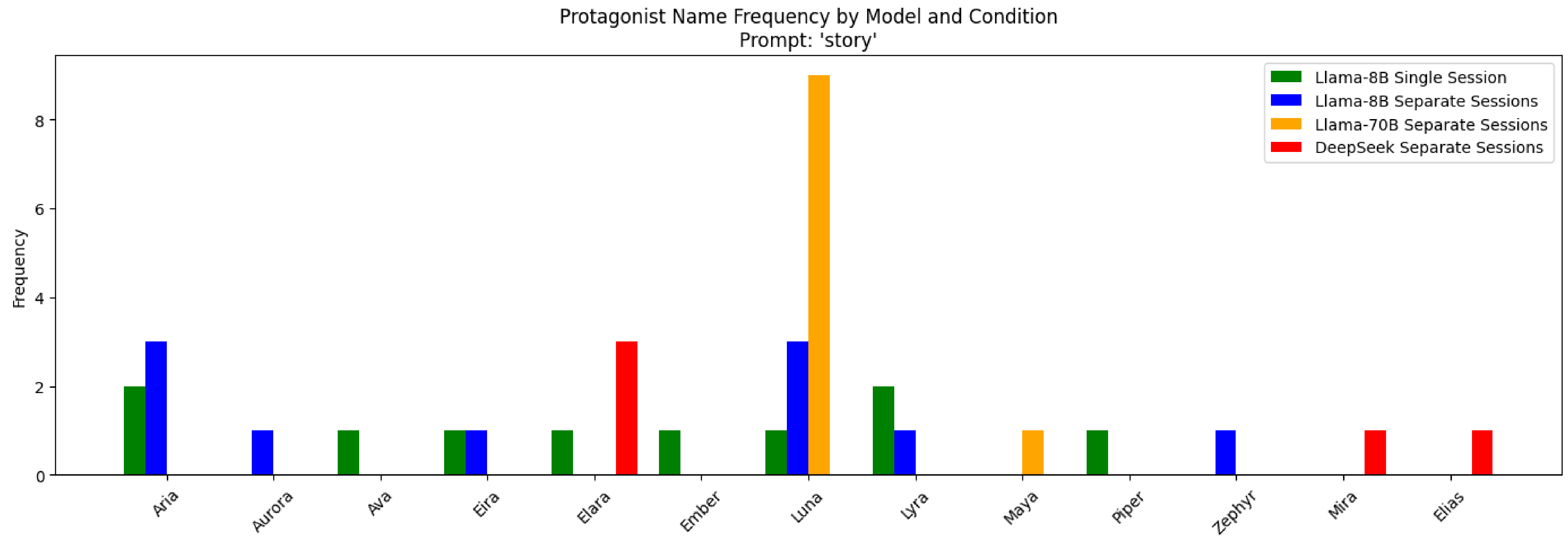


Prototype Completeness per model





Name Frequency per model



What we can draw from this

- Prototype completeness depends on three factors:
  - Model scale, context and training corpus
- Context stabilizes output:
  - Single Session with prompt “story” leads to more coherent outputs
- Explicit prompts lead to more coherent outputs when no prior context is given
- Scale affects prototype completeness as well
- Different models have different default stories (Llama vs. DeepSeek)

Evaluation of Benzon’s Paper

Based on the Experiment’s Main Findings

Claim 1: Stories in Single Session more variety than in Separate Sessions based on name “Elara”

Knowing that ChatGPT takes the entire session to date into account when it responds to a prompt, I assumed that generating many stories in a single session would have some effect of the stories. I conjectured that there would be greater variety among the 10 stories generated in the single session than among the 10 stories where each was generated in its own session. A quick look at the protagonists in the 20 stories suggests that this is true. Two out of the first 10 stories have a protagonist named “Elara,” while five out of the second set of 10 are named “Elara.” That quick and dirty assessment, however, needs to be confirmed by further analysis.

- Single stories make the models outputs more consistent and coherent
- The model sees its own stories and copies the pattern reliably
- However, the results of the experiments also show that the Single Session experiment with the prompt “story” has a higher name variety than outputs in the Separate Session experiment
- The stories in the single session are varied on the surface (names, settings,..) but the overall storyline stays the same

Claim 2: prototypical story of ChatGPT is abstraction over all the stories it encountered during training

Given that that one-word prompt, "story," places no restrictions on ChatGPT, why does ChatGPT return to that one kind of story? In the absence of any further information, I have to assume that kind of story is an abstraction over all the stories that ChatGPT has encountered during training. That, in effect, is ChatGPT's prototypical story. Think about that for a moment. The large language model (LLM) at the core of ChatGPT was created by a GPT engine given the task of predicting the next word in an input string. Those strings come from millions of texts in a corpus that embodies a large percentage of the texts that are publicly available on the world-wide web. The prototypical story is the model's response to that single word, story, and thus somehow embodies predictive information about, well, stories. The fact that ChatGPT generates that particular generic story in response to the prompt, story, is a clue about the structure of the underlying model.

- DeepSeek proves this directly: It tells complete different stories (different storylines and elements, no named protagonists) because it was trained on different data
- The fairy tale prototype isn´t universal, it is specific to models trained heavily on Western fantasy fiction
- Prototype: Gives us an assumption about the model's training corpus

Claim 3: Studying the outputs of models is just as important as studying its internal mechanistics

To belabor the example and put it to use as an analogy for mechanistic interpretability, someone with a good feel for mechanisms can tell you a great deal about how the parts of this strange device articulate, the range of motion for each part, the stresses operating on each joint, the amount of force require to move the parts, and so forth. But, still, when you put all that together, that will not tell you what the device was designed to do. The same is true, I believe, for the mechanistic investigation of Large Language Models (and other types of neural net models). These models are quite different from mechanical devices such as olive-pitters or internal combustion engines. Their parts are informatic, bits and bytes, rather than physical objects, and there are many more of them, "billions and billions" as Karl Sagan used to say. Moreover, olive-pitters and internal combustion engines were designed by human engineers while LLMs were not designed by anyone. Rather, they evolved through a process whereby a software engine, designed by engineers and scientists, "consumed" a large corpus of texts. The very strangeness of the resulting model, its "black box" nature, makes it imperative that we investigate the structure of the texts that it produces.

- The experiments also support this
- **Blackbox approach:** By looking at the outputs of different models, one can tell how the model was probably trained and how it generates text
- The prototypical story acts like a diagnostic tool

Connections to the Paper "ChatGPT tells stories, and a note about reverse engineering: A Working Paper, Version 3"

- Paper about how ChatGPT adjusts stories based on the protagonist's nature
  - William the lazy gives his knights a potion to make them immune to dragon's flames -> knights were able to defeat the dragon
  - Henry the eloquent uses his trait to calm the dragon down with words
  - Connection: Prototypical protagonist usually nature loving and curious woman and her surrounding (enchanted forests, wise trees, ..) matches her traits
- Stories have five segments: Donné, Disturb, Plan/Transit, Enact and Celebrate
  - Connection: Prototypical story line Home, Departure, Journey/Quest, magical world and wisdom, return and community benefit
- **Overall:** both papers investigate the coherence of stories ChatGPT generates, what its default story is and how it reacts to changes in the traits of protagonist

A Model-Based Framework for Developing Certifiable Autonomous Cognitive Systems: Driving as a Case in Point

Prof. Dr. Visvanathan Ramesh and Prof. Dov Dori

Professor Ramesh's and Dori's work on vision systems

- **Problem:** How can we prove certifiability of autonomous driving?
- **Certifiability:** autonomous systems (especially self-driving cars) should be at least as safe as an expert human driver
- Approach with two tools:
  - **Top-down modeling (Dori):** Using OPM (modeling language) to specify what an autonomous system does, structure, behavior, all in one diagram type
  - **Bottom up engineering/testing (Ramesh):** Software platform for rapidly building and testing performance of computer vision components on different settings

Performance characterization

- How does a vision/sensor component's performance change based on its inputs? (weather, lighting, noise, algorithms,..)
- **Goal:** predict when/how a component will degrade, before it fails in the real world
- **Tools for performance evaluation of a component:**
  - **Blackbox:** only look at inputs and outputs of component
  - **Glassbox:** blackbox + inspecting inner workings of algorithm
  - **Whitebox**: full mathematical description of transform, input- and output distributions
- **Methods used to classify performance:**
  - **Analytical:** mathematically derive how input noise determines an algorithms output to predict output error
  - **Empirical:** synthetic, parametric graphics-based simulations for generating traffic scenes with known ground truth and running vision systems on them

Connection to Benzon’s work

- **Treats ChatGPT as a blackbox**
  - No internal weights/architecture
  - Only inputs (prompts) and outputs (stories)
- **Empirical performance characterization:**
  - Same prompt (“story”), 10x in single session vs. 10x separate session
  - Statistically observe patterns in output
- **Overall parallel:**
  - Same blackbox/empirical logic
  - Repeated trials to characterize system behavior
  - Core difference: LLM vs. vision system

Thanks for listening!